The Anchor Protocol:

A Mechanically Analyzed Control Plane
for Local Agent Swarms

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Abstract

As AI agents transition from isolated copilots to collaborative, autonomous swarms, local development environments face a crisis of coordination and trust. Existing tools designed for human-driven, static infrastructures lack the primitives required to prevent dynamic agents from corrupting shared state or squatting on ephemeral ports.

This paper introduces the **Anchor Protocol**, a cryptographic and semantic identity framework built into the Port Daddy daemon. We detail the protocol's evolution from symmetric MACs to Ed25519 signatures and multi-hop delegation chains inspired by Macaroons [4]. ProVerif [1] checks symbolic authentication and attenuation properties of the protocol models; Kani checks bounded memory-safety and source-level control-flow properties of the Rust core; runtime conformance tests cover the deployed bridge. These layers do not amount to a proof of the entire implementation or of hardware-level constant time. The multi-hop proof is also structurally bounded, distinct from ProVerif's unbounded-session quantification. Section 7.3 records those boundaries explicitly.

Keywords: distributed systems security, formal verification, capability-based security, multi-agent coordination, symbolic protocol analysis, ProVerif, JWT security, Ed25519

Reading time: approximately 30 minutes end-to-end. Chapter VI, The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems *(Owens 2026), carries the economic and governance argument; either paper can be read first.*

PORT DADDY COORDINATION PAPERS **CHAPTER V OF VII**

V. The Anchor Protocol (this chapter). Authentication, delegation, attenuation, and revocation inside one harbor.

VI. The Bonded Commons. Evidence, bonding, advisory claims, and the incentive conditions for cooperation.

VII. The Federated Harbor. Identity, revocation, and settlement across mutually sovereign machines.

Chapters I–IV form the product arc; Chapters V–VII provide the protocol, economic, and federation deep dives. Every chapter is independently readable.

Contents

1 Introduction: The "Local Swarm" Problem

In a multi-agent development environment, agents operate concurrently to read files, spin up test servers, and execute commands. This introduces three critical threat vectors on `localhost`:

1. **Port Squatting (The "Ghost in the Harbor"):** If Agent A crashes, a malicious or malfunctioning Agent B can bind to Agent A's previously assigned port, intercepting traffic meant for A.

2. **Resource Contention:** Agents fighting over shared resources (e.g., a SQLite database file) without a centralized locking mechanism cause state corruption.

3. **Privilege Escalation:** A "Reviewer" agent delegated by a "Coder" agent should not possess the rights to initiate production deployments, yet local environments rarely enforce principle-of-least-privilege between processes.

To solve this, Port Daddy introduces the concept of a **Harbor**—a zero-trust, namespace-isolated execution environment where agents must prove their identity and capabilities. Our approach builds upon foundational work in capability-based security by Dennis and Van Horn [7] and the principle of least privilege articulated by Saltzer and Schroeder [15].

Volume Context. This chapter provides the cryptographic substrate. Chapter VI (*The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems*) builds the economic and game-theoretic layer on top: bonded participation, conditional auctions, mechanism analysis, and Sen-inspired advisory conflict resolution. The Anchor Protocol is a self-contained subset consumed as the trust kernel for the wider mechanism design. Read this chapter for "how does the daemon authenticate an agent?"; read Chapter VI for "why should there be a daemon at all?"

2 The Anchor Protocol Architecture

The Anchor Protocol defines how agents authenticate to a Harbor and to each other. It issues **Harbor Cards** (highly constrained JSON Web Tokens) and evolves them across four phases, each adding a capability that closes a specific attack surface from the previous phase (Figure 1).

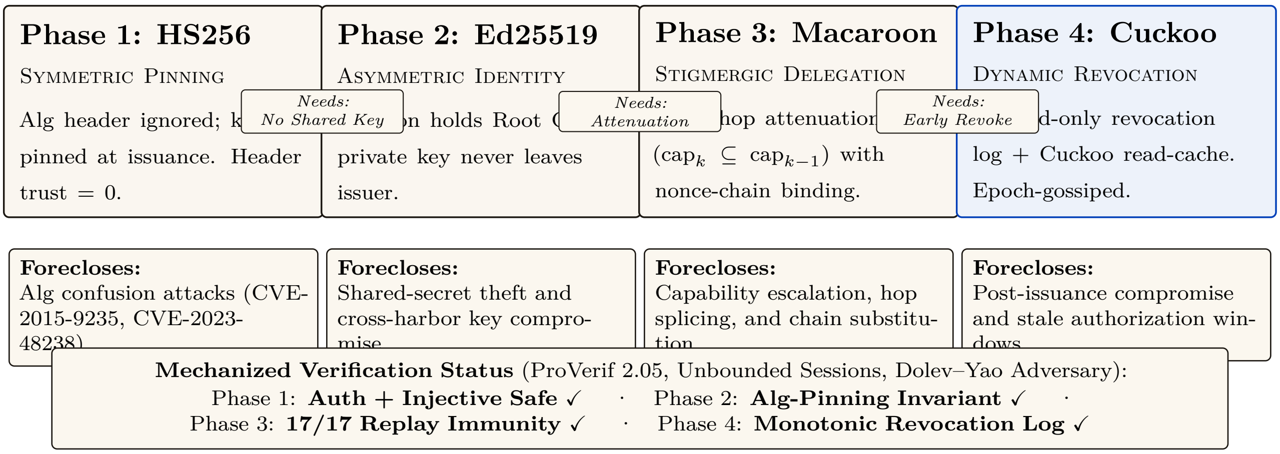


Figure 1: The four phases of the Anchor Protocol. Each phase refines the previous: the cryptographic primitive strengthens, and a specific adversary threat is formally foreclosed. Phases 1–3 are mechanized in ProVerif (§A); Phase 4 is enforced via append-only gossip logs and monotonic filter proofs (§2.4).

A common thread across all phases is *capability attenuation*: a delegated token never carries more authority than its parent, and TTL only shrinks. Figure 2 shows the nested-cap structure that the verifier enforces on every chain.

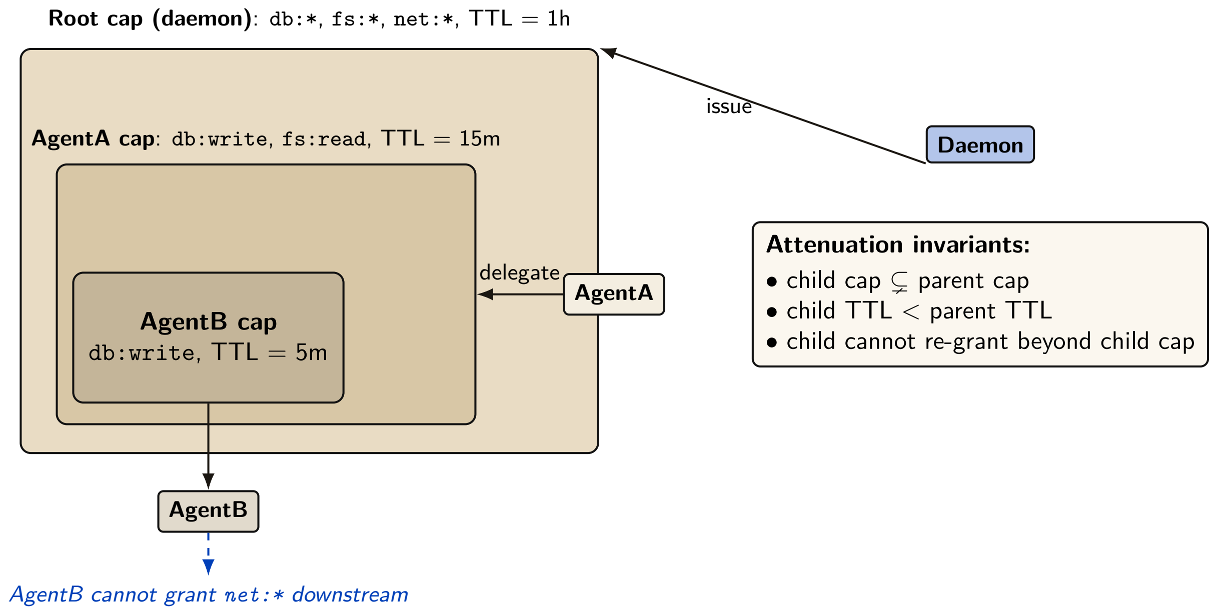


Figure 2: Capability attenuation under delegation. Each child capability is a strict subset of its parent's (rights and TTL both shrink monotonically). AgentB inherits only what AgentA explicitly delegates, and cannot re-grant capabilities it does not hold. This is enforced both at issuance (parent signs over the child's attenuated claim set) and at verification (the verifier checks the chain of `is_subset` relations).

2.1 Phase 1: Symmetric Pinning

Early versions of the protocol relied on HS256 (HMAC-SHA256). The primary vulnerability in JWTs is "Algorithm Confusion" (CVE-2015-9235 [11], CVE-2023-48238 [13]), where an attacker modifies the token header to `{"alg": "none"}` or symmetric HS256 while using an asymmetric public key as the HMAC secret.

Property 2.1 (Algorithmic Pinning)**.** The Port Daddy Verifier employs strict **Algorithmic Pinning**. It entirely ignores the user-provided `alg` header and explicitly forces the underlying crypto library to evaluate the signature using the pre-determined algorithm. This approach is recommended by the JWT Best Current Practices draft [14].

Figure 3 shows the two verifier paths side by side: the naive verifier trusts the `alg` field on the wire (and is exploited), while the pinned verifier records the issuance algorithm out-of-band and admits no rewrite trace.

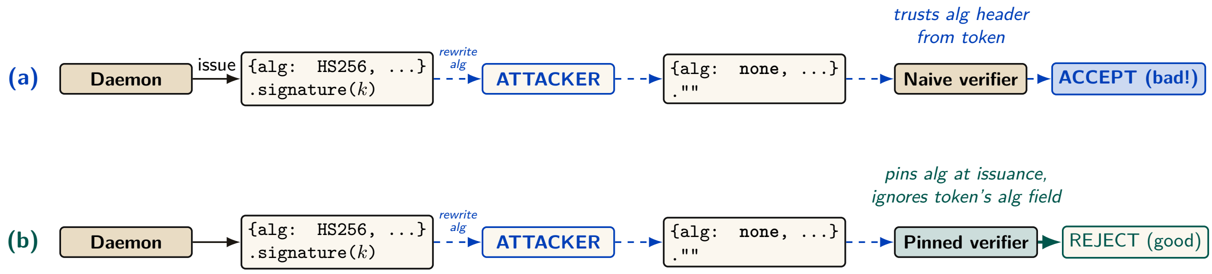


Figure 3: Algorithm-confusion attack and the pinned-verifier defense. (a) A naive verifier trusts the `alg` field of the incoming token; an attacker who can intercept the token rewrites `alg=HS256` to `alg=none`, strips the signature, and the verifier accepts an unauthenticated message. (b) The pinned verifier records the issuance algorithm out-of-band, then re-checks each token against the pinned algorithm rather than the token's self-described one. The ProVerif model encodes both verifiers and shows the naive path produces a counter-trace witness for the rewrite, while the pinned path admits no such trace. Runtime conformance: 5 attack patterns + 2 controls, all pass.

2.2 Phase 2: Asymmetric Identity (Ed25519)

To support decentralized Agent-to-Agent (A2A) communication without sharing HMAC secrets, the protocol transitioned to **Ed25519 (EdDSA)** [8, 9]. Ed25519 provides fast, deterministic signatures with small keys and signatures, making it ideal for local agent communication.

Definition 2.1 (Harbor Key Hierarchy)**.** The Port Daddy Daemon acts as the Root Certificate Authority (CA). Each Harbor is assigned a unique Ed25519 keypair. The Daemon issues Harbor Cards signed by the Harbor's private key. Agents use the Harbor's public key (broadcast via Port Daddy's ambient discovery service) to verify tokens presented by peers.

2.3 Phase 3: Stigmergic Delegation (The "Biscuit" Pattern)

Port Daddy v4 introduces multi-hop delegation. When Agent A spawns Agent B, it does not need to request a new token from the Daemon. Instead, it uses **Offline Attenuation**, inspired by Macaroons [4].

Definition 2.2 (Offline Attenuation)**.** Agent A takes its existing Harbor Card, appends a new set of restricted capabilities (e.g., `["db:read"]` reduced from `["db:read", "db:write"]`), and signs the *entire chain* with its own ephemeral private key.

The Harbor verifies the Daemon's root signature, Agent A's delegation signature, and mathematically ensures that Agent B's capabilities are a strict subset of Agent A's.

For depth-N chains, the verifier walks the structure shown in Figure 4. Each hop must produce a fresh nonce and bind the previous and next identities into its signature so that an attacker who intercepts a single hop cannot splice it into a different chain.

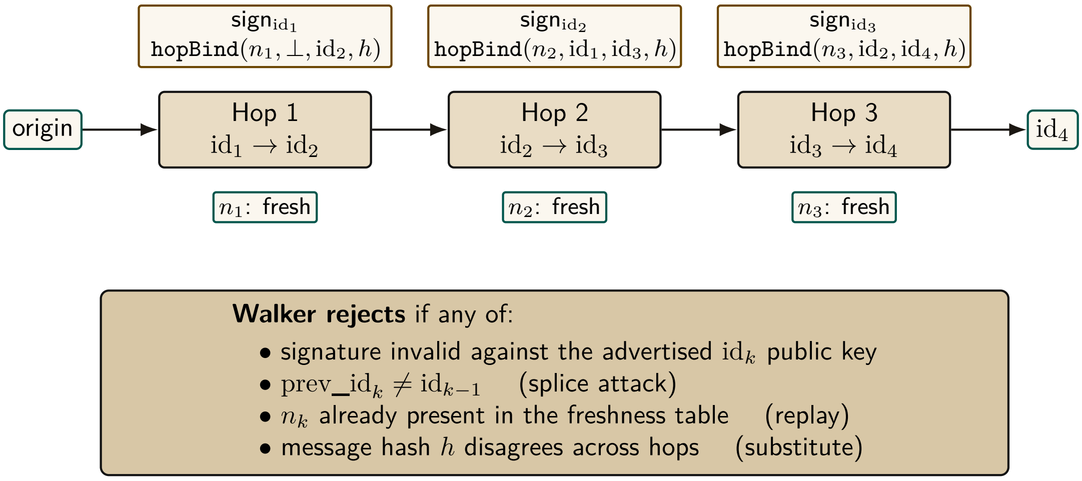


Figure 4: Multi-hop delegation walker. Each hop k signs the binding tuple $(n_k, \mathrm{id}_{k-1}, \mathrm{id}_k, h)$ under Ed25519, where h is the message hash that must remain constant across the chain. The walker verifies depth-N chains with a nonce-freshness table and rejects splices, replays, hop-swaps, and substitution. Runtime: `lib/delegation-chain.ts`; ProVerif conformance test passes 17 / 17.

2.4 Phase 4: Revocation via Cuckoo Filter

Capability tokens have a TTL, but natural expiry is not always sufficient. Two circumstances force early withdrawal: (i) *compromise*—a Harbor Card leaks from its agent's process, and every second until TTL is a second the attacker has legitimate capabilities; and (ii) *policy reversal*—a principal decides an agent should no longer hold `db:write`, and waiting for TTL is unacceptable. A naive revocation list is $O(n)$ per verification and grows monotonically. While the Anchor Protocol previously attempted to gossip a cuckoo filter directly (which succumbed to the bitwise merging impossibility), it now gossips an **Append-Only Event Log of Revocation IDs** using Vector Clocks. To maintain $O(1)$ read performance, each daemon continually projects its event log into a local, ephemeral cuckoo filter [25] (Figure 5).

Cuckoo filters in one paragraph. A cuckoo filter is a compact probabilistic set: under successful insertion and correct deletion discipline it has false positives but no false negatives, while supporting deletion. Each inserted key is reduced to a fingerprint and placed in one of two candidate buckets; on collision, fingerprints are relocated until a slot opens or insertion fails. Lookups inspect both buckets. The expected lookup cost is $O(1)$ and the approximate false-positive probability for two buckets of size B and f fingerprint bits is at most about $2B/2^f$ in the low-collision model. High load increases insertion failures; the implementation rejects or rebuilds rather than silently dropping a revocation. Duplicate fingerprints require counted or authoritative-table-backed deletion so that removing one key does not reanimate another.

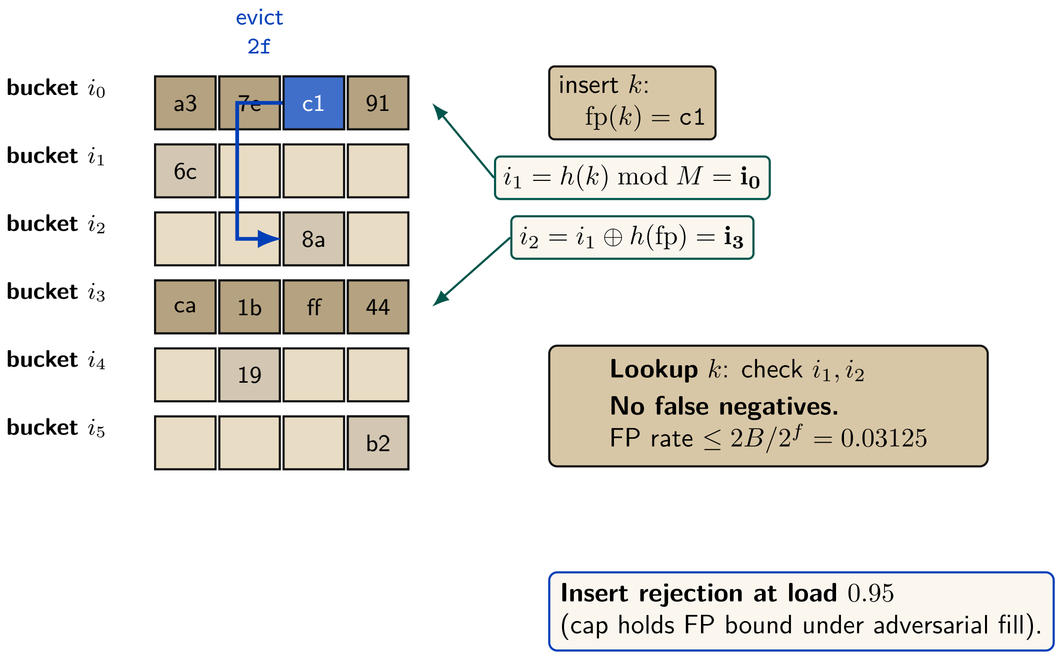


Figure 5: Cuckoo-filter revocation gate (Fan et al. [25]) with the load-factor ceiling that defeats the pollution attack. Insert lands at either of two candidate buckets $i_1, i_2 = i_1 \oplus h(\mathrm{fp})$; on full, evict and re-insert at the kicked fingerprint's alternate. Insert rejection at 0.95 caps occupancy inside the regime where the false-positive bound holds.

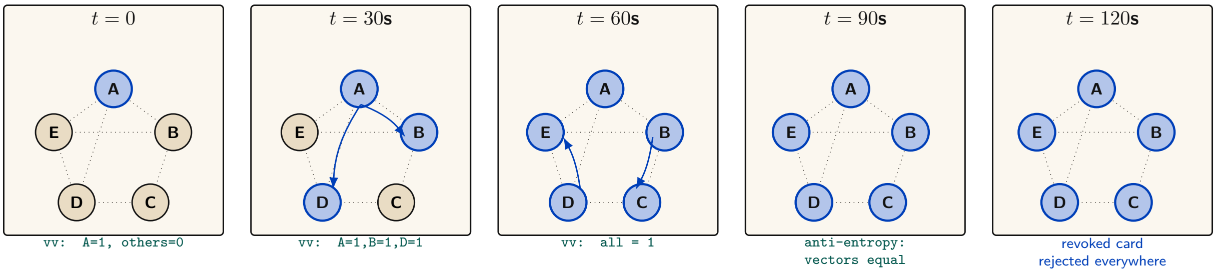
Why cuckoo, not Bloom. For uniform-TTL workloads, a rotating Bloom filter would suffice. Port Daddy's TTLs are heterogeneous—Shipwright proposals run hours to days, `pd spawn` children run minutes, one-shot agents run seconds. Rotating Bloom either rotates at the longest TTL (short-TTL revocations linger orders of magnitude longer than needed) or buckets by TTL class (reintroducing the structural complexity cuckoo collapses). Cuckoo provides $O(1)$ per-entry removal at each specific expiry, plus the same removal handles operator-initiated early-revocation undo cleanly.

Space. At false-positive rate ϵ, a cuckoo filter uses approximately $\log_2(1/\epsilon) + 3$ bits per entry. For $\epsilon = 10^{-3}$, ≈ 13 bits per revoked card. A fleet retaining 10^5 active revocations fits in $\approx 160\,\mathrm{KB}$—trivially in memory, trivially gossiped.

Definition 2.3 (Revocation-Augmented Verification)**.** Let R_t denote the authoritative Append-Only Event Log of revoked Harbor Card IDs at time t. Each daemon d maintains a local, ephemeral cuckoo filter index $F_d \approx R_t$. Every Harbor Card verification additionally:

1. Looks up the card's `kid` in F_d. If absent, admit.
2. If present, query the local `revocations` table (authoritative). If genuinely revoked, reject with `revoked`; otherwise the filter is in the false-positive regime and the caller is admitted, with an asynchronous reconciliation job repopulating the filter.

Gossip-based synchronization. Daemons in a mesh synchronize revocation deltas by anti-entropy gossip [26]. Under a connected complete-overlay model with independent uniform peer selection and reliable rounds, epidemic dissemination is expected $\Theta(\log m)$ rounds. This is an expectation, not a deadline; partitions, message loss, topology, and scheduling change the constant or prevent global convergence until the partition heals. Security-critical revocations can additionally trigger best-effort immediate push. Figure 6 illustrates propagation across five daemons; it is a trace, not a timing proof.



Mechanism. Each daemon maintains a cuckoo filter (8-bit fingerprints, 4 entries/bucket, FP rate $\leq 2^{-5}$, load ceiling 0.95) and a *version vector* over its peers. Every 30 s each daemon picks a random peer, exchanges vectors, and ships deltas (cards revoked since the peer's last-seen version).

Expected propagation. In a connected reliable-round random-peer model, a new revocation reaches all m peers in expected $\Theta(\log m)$ rounds. The pictured 30 s steps are an illustrative trace, not a deadline; partitions admit no finite global bound.

Legend. sand-filled node = filter clean · cobalt-ringed node = filter contains revoked card · $\longrightarrow$ = gossip delta this round

Figure 6: Gossip-synchronized revocation across a five-daemon mesh. At $t = 0$ daemon A locally revokes a Harbor Card (its filter and revocation table). Anti-entropy gossip ships deltas to random peers every 30 s. Expected $\Theta(\log m)$ propagation requires the stated connected reliable-round model; the panels show one possible execution, not a worst-case time guarantee. Version vectors make duplicate and reordered deltas idempotent once delivered, but do not heal an unbounded partition.

Property 2.2 (Filter Monotonicity over a TTL Epoch)**.** For any Harbor Card c with issue time t_0 and TTL τ, if $c \in R_t$ for some $t \in [t_0, t_0 + \tau]$, then $c \in R_{t'}$ for all $t' \in [t, t_0 + \tau]$. After $t_0 + \tau$, c may be removed from R since the card has expired.

Property 2.3 (No Partial Reanimation)**.** A revoked card cannot be un-revoked within its TTL. If an operator decides the revocation was erroneous, they must issue a new card.

Property 2.4 (Conditional Gossip Expectation)**.** Under the complete-overlay, uniform-peer, reliable-round model above, a revocation disseminates to all m daemons in expected $\Theta(\Delta \log m)$ time. No finite worst-case freshness bound holds under an unbounded partition.

Soundness preservation. Revocation strictly narrows acceptance, so existing ProVerif soundness proofs (Theorem 4.1) are unaffected. The new proof obligation is small: revocation is *enforced*, which follows by inspection of the verification path—the revocation check is unconditional and precedes admission. The card lifecycle gains a state (*valid* $\to$ *revoked* $\to$ *expired*) modeled as an additional transition.

Privacy. A Dolev-Yao adversary observing all gossip traffic learns the revoked-card IDs, leaking which agents have been disciplined. For an organization's own daemons, this is acceptable audit transparency. Stronger privacy is available by encoding revocations as salted hashes $H(\texttt{kid}, \texttt{salt})$ where the salt rotates daily, stored encrypted under a harbor session key.

3 Related Work

3.1 Formal Verification of Security Protocols

Formal verification of cryptographic protocols has matured significantly over the past two decades. Following the seminal work by Lowe [5] on authentication hierarchies and the Dolev-Yao model [6], several automated tools have been developed:

- **ProVerif** [1]: An automatic symbolic protocol verifier supporting an unbounded number of sessions, used to verify TLS 1.3 [17], Signal [16], and WireGuard.
- **Tamarin** [18]: A state-of-the-art protocol verifier that supports equational properties and inductive proofs, used for analyzing 5G authentication [19].

- **CryptoVerif** [2]: A computationally-sound protocol verifier that provides proofs in the computational model rather than the symbolic model.

Our work builds on these foundations, applying established verification techniques to the novel domain of local multi-agent coordination.

3.2 Capability-Based Authorization

Capability-based security dates back to Dennis and Van Horn's seminal 1966 paper on programming semantics for multiprogrammed computations [7]. Recent work by Birgisson et al. [4] introduced Macaroons, which use chained HMACs for decentralized delegation with contextual caveats. The Anchor Protocol's delegation chains (Section 5.3) are inspired by Macaroons but adapted for JWT-based identity in local development environments.

3.3 JWT Security

JSON Web Tokens have been the subject of extensive security analysis. Algorithm confusion attacks were first systematically studied by McLean [10], with subsequent vulnerabilities documented in CVE-2015-9235 [11] (HS256/RS256 confusion), CVE-2018-1000531 [12] ("none" algorithm), and CVE-2023-48238 [13] (generic algorithm confusion). Our protocol implements the defense recommended by the IETF JWT Best Current Practices [14]: strict algorithm whitelisting and key separation.

3.4 Credential Formats in Agentic Commerce

Since this protocol was first specified, the agentic-payments industry has converged on a concrete credential dialect: the Agent Payments Protocol [28], adopted by the Universal Commerce Protocol [27] (Google/Shopify, 2026), carries its authorization mandates as W3C Verifiable Credentials in SD-JWT form — SD-JWT+kb for principal proof, JWS detached-content signatures (RFC 7515 Appendix F) over JCS-canonicalized payloads (RFC 8785) for counterparty authorization, with ES256 as the recommended algorithm and keys published as JWK sets in a `/.well-known` profile. The reference implementation migrates the Harbor Card envelope to this profile (`hv: 3`, ADR-0094) so that external verifiers need no bespoke SDK. The migration is deliberately envelope-only: the delegation-chain semantics, the attenuation invariant of Theorem 4.1, and the ProVerif obligations are independent of the serialization, though the key-binding construction of SD-JWT+kb adds a binding obligation the model must re-discharge. The JWT hardening above (strict algorithm whitelisting, key separation) transfers unchanged — SD-JWT is a JWT profile, not a departure from one.

4 Formal Verification Strategy

Following the industry standard set by AWS (s2n-tls [20]) and Microsoft (Project Everest [22]), we decoupled the verification of the *protocol design* from the verification of the *implementation*. The full stack has three layers (Figure 7): symbolic protocol analysis at the top, bounded model checking on the Rust source in the middle, and runtime conformance tests against the deployed binary at the bottom. Each layer's obligations flow downward; each layer's evidence flows back up. The stack is sound only if every bridge — symbolic-to-source and source-to-deployed — holds.

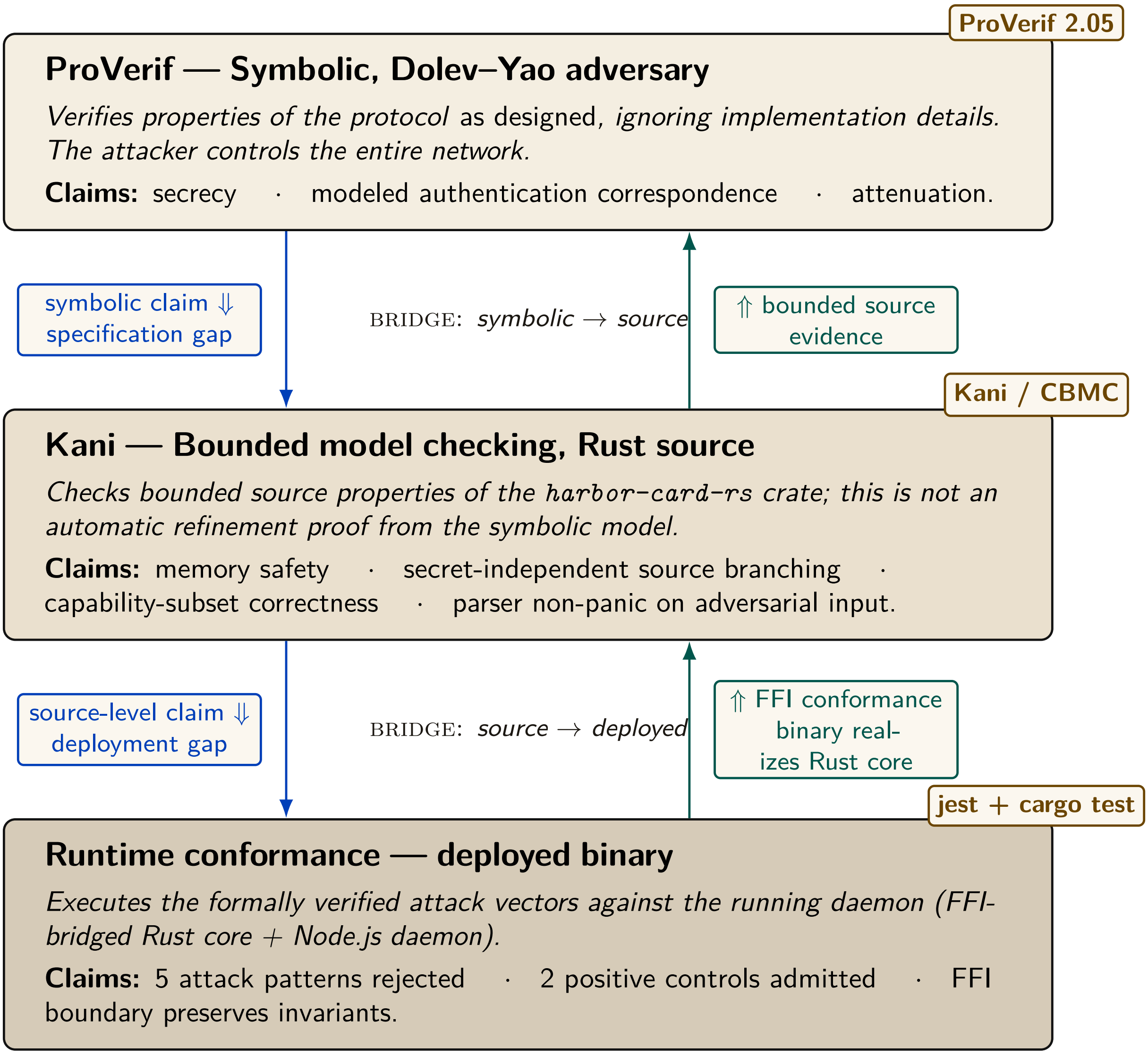


Figure 7: The three-layer verification stack. ProVerif (top) discharges design-level obligations against a Dolev–Yao adversary. Kani (middle) checks bounded Rust source-level properties that overlap with, but do not refine the full symbolic model. Runtime conformance tests (bottom) exercise the FFI-bridged binary with selected attack vectors; they reduce rather than eliminate the deployment gap. Cobalt arrows show obligations flowing down (what each lower layer must preserve); teal arrows show evidence flowing back up. The unproved bridges remain explicit assurance-case assumptions rather than inferred theorems.

4.1 Symbolic Analysis with ProVerif

To prove the protocol's logical soundness against a Dolev-Yao adversary [6] (an attacker who controls the entire network), we modeled the Anchor Protocol in **ProVerif** [1]. ProVerif represents protocols as Horn clauses and uses resolution to prove security properties for an unbounded number of sessions.

Theorem 4.1 (Authentication Correspondence in the Modeled Phase)**.** For the modeled delegation phase, ProVerif confirmed the correspondence:

$$\begin{aligned} \mathrm{Accepted}(b, h, cap) \implies\ & (\mathrm{IssuedRoot}(a, h, cap) \wedge \mathrm{Delegated}(a, b, cap)) \\ & \vee\ \mathrm{IssuedRoot}(b, h, cap) \end{aligned} \tag{1}$$

The displayed query is a non-injective authentication correspondence: every accepted event has a matching modeled issuance/delegation event. It does not by itself prove replay freedom, capability attenuation, or arbitrary-depth transitivity; those are separate queries and bounds.

All three protocol phases were independently verified in ProVerif 2.05. Table 1 summarizes the results; full models are reproduced in Appendices B, C, and D.

Query	**Phase**	**Model**	**Result**
$\neg$ attacker($master\_key$)	v1 (HS256)	`v1_refined.pv`	**TRUE**
Accepted $\implies$ Issued	v1 (HS256)	`v1_refined.pv`	**TRUE**
$\neg$ attacker($harbor\_sk$)	v2 (Ed25519)	`v2_asymmetric.pv`	**TRUE**
Accepted $\implies$ Issued	v2 (Ed25519)	`v2_asymmetric.pv`	**TRUE**
Accepted(b) $\implies$ IssuedRoot(a) $\wedge$ Delegated(a, b)	v3 (Delegation)	`v3_delegation.pv`	**TRUE**

Table 1: ProVerif results for the listed model queries. Session replication is unbounded where the model uses replication; delegation-chain structure remains bounded as stated in §7.3.

The following is the verbatim ProVerif 2.05 output for the delegation chain query (Phase 3), the most complex property:

```
RESULT event(Accepted(b_2,harbor_id[],cap_1))
  ==> (event(IssuedRoot(a_3,harbor_id[],cap_1))
       && event(Delegated(a_3,b_2,cap_1)))
      || event(IssuedRoot(b_2,harbor_id[],cap_1))
  is true.
```

Listing 1: ProVerif 2.05 Terminal Output — Phase 3 Delegation Chain

4.2 Runtime Enforcement: The Arbiter

Formal models establish properties of their abstractions. At runtime, Port Daddy deploys the **Arbiter**—an ambient monitor that subscribes to the daemon's post-commit activity log and checks six derived invariants:

1. `PID_SQUATTING`: Detects when a process claims a port previously held by a different PID (the "Ghost in the Harbor").
2. `CAP_ESCALATION`: Verifies capability subsets via the deployed Rust FFI enforcer.
3. `NOTE_MONOTONICITY`: Ensures the note sequence for active sessions never shrinks (from the TLA$^+$ `NoteMonotonicity` invariant).
4. `ESCROW_POSITIVE`: Validates that active sessions have positive escrow (when Float Plans are active).
5. `LOCK_OWNER_VALID`: Ensures locks are only held by registered agents with active sessions.
6. `HEARTBEAT_FRESHNESS`: Flags stale heartbeats as early warning for agent death.

Violations are logged and can trigger the "man overboard" salvage protocol. Because this path observes committed events, it detects and compensates; only a pre-commit native guard can prevent a transition. The Arbiter API provides visibility, not a proof that every security-relevant operation was mediated.

4.3 Implementation Verification

A secure protocol is useless if the implementation contains buffer overflows or timing leaks. We extracted the critical JWT parsing and cryptographic comparison logic into a verified core.

Property 4.1 (Memory Safety)**.** Using the **Kani** Rust Verifier [21], we performed bounded model checking to prove that our parsing logic never panics or accesses out-of-bounds memory, regardless of the input payload.

Property 4.2 (Secret-Independent Source Control Flow)**.** The Rust comparator uses no source-level branch on secret byte values, and Kani checks that bounded control-flow property. This is useful evidence but not a hardware-level constant-time proof: compiler transformations, memory behavior, and microarchitecture remain outside the model.

5 Verification Algorithms

In this section, we present the verification algorithms in pseudocode format, following the notation used in protocol verification literature [1, 5]. These algorithms correspond to the formal ProVerif models in Appendix B.

5.1 Phase 1: Symmetric HS256 with Algorithm Pinning

Algorithm 2 presents the secure verification procedure that is immune to algorithm confusion attacks (CVE-2015-9235 [11]).

```
1  Require: Pinned Algorithm: hs256
2  Require: Master Key: k_master (secret)
3
4  function Daemon.IssueToken(agent_id, harbor_id, capability):
5      jti <- GenerateNonce()
6      msg <- Encode(agent_id, harbor_id, jti, capability)
7      sigma <- HMAC-SHA256(msg, k_master)
8      emit Issued(agent_id, harbor_id, jti, capability)
9      return (hs256, msg, sigma)
10
11 function Harbor.VerifyToken(tok, expected_harbor):
12     (alg_header, msg, sigma) <- tok
13     // CRITICAL: Ignore alg_header, pin to HS256
14     if HMAC-SHA256(msg, k_master) != sigma:
15         return _|_  // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_  // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)
```

Listing 2: Algorithm 1: Secure Harbor Verification (HS256 with Algorithm Pinning)

Security Property: The algorithm ignores the user-provided alg_{header} (line 15), preventing algorithm confusion attacks where an attacker sets alg = "none" or attempts HS256/RS256 confusion [10].

5.2 Phase 2: Asymmetric Ed25519

Algorithm 3 presents the Ed25519-based verification. This phase removes the need for shared HMAC secrets between distributed Harbors.

```
Require: Harbor Keypair: (sk_harbor, pk_harbor)
Require: Key Distribution Channel: c_key (private)

function Daemon.IssueToken(agent_id, harbor_id, capability):
    sk_h <- Receive(c_key)  // Get harbor's secret key
    jti <- GenerateNonce()
    msg <- Encode(agent_id, harbor_id, jti, capability)
    sigma <- Ed25519.Sign(msg, sk_h)
    emit Issued(agent_id, harbor_id, jti, capability)
    return (msg, sigma)

function Harbor.VerifyToken(tok, pk_h, expected_harbor):
    (msg, sigma) <- tok
    if not Ed25519.Verify(msg, sigma, pk_h):
        return _|_  // Invalid signature
    (a, h, j, cap) <- Decode(msg)
    if h != expected_harbor:
        return _|_  // Wrong harbor
    emit Accepted(a, h, j, cap)
    return (a, h, j, cap)
```

Listing 3: Algorithm 2: Asymmetric Harbor Verification (Ed25519)

Security Property: The secrecy of sk_{harbor} and the unforgeability of Ed25519 [8] ensure that only the legitimate Daemon can issue valid Harbor Cards.

5.3 Phase 3: Multi-hop Delegation

Algorithm 4 presents the delegation chain verification, inspired by Macaroons [4].

```
Require: Daemon Public Key: pk_daemon
Require: Subset Relation: subseteq on capabilities

function Agent.Delegate(token_parent, sk_parent, agent_child, cap_sub
    ):
    (msg_parent, sigma_parent) <- token_parent
    (_, agent_parent, harbor, cap_parent) <- Decode(msg_parent)
    if cap_sub not_subseteq cap_parent:
        return _|_  // Cannot escalate capabilities
    msg_delegate <- Encode_delegation(msg_parent, sigma_parent,
        agent_child, cap_sub)
    sigma_delegate <- Ed25519.Sign(msg_delegate, sk_parent)
    emit Delegated(agent_parent, agent_child, cap_sub)
    return (msg_delegate, sigma_delegate)

function Harbor.VerifyChain(chain, pk_root):
    tok_root <- chain[0]
    (msg_0, sigma_0) <- tok_root
    if not Ed25519.Verify(msg_0, sigma_0, pk_root):
        return _|_  // Invalid root signature
    cap_prev <- ExtractCaps(msg_0)
    pk_current <- pk_root
    for i <- 1 to |chain| - 1:
        (msg_i, sigma_i) <- chain[i]
        if not Ed25519.Verify(msg_i, sigma_i, pk_current):
            return _|_  // Invalid delegation signature
        cap_i <- ExtractCaps(msg_i)
        if cap_i not_subseteq cap_prev:
```

```
            return _|_  // Per-hop capability escalation detected
        cap_prev <- cap_i
        pk_current <- ExtractPubkey(msg_i)
    emit Accepted(agent_final, harbor, cap_prev)
    return T
```

Listing 4: Algorithm 3: Delegation Chain Verification

Security Property: Monotonic per-hop attenuation ($\mathrm{cap}_i \subseteq \mathrm{cap}_{i-1}$) prevents capability expansion across transitive delegations (preventing `write` → `read` → `write` re-escalation). Theorem D.1 (§D.1) proves inductive soundness under the partial order.

6 Implementation: Deployed Rust Core

The verified cryptographic core is not a reference implementation—it is the **deployed production code**. The open-source Rust crate `harbor-card-rs` is compiled to a C dynamic library (`cdylib`) and called from the Node.js daemon via FFI through the daemon's runtime-enforcement (Arbiter) module.

This architecture eliminates the gap between "verified code" and "running code" that weakens many formal verification efforts. The same binary that Kani model-checks is the same binary that the daemon loads at runtime.

6.1 Core Verification Logic

```
pub fn verify(&self, token: &str, now_ts: i64)
    -> Result<HarborCardClaims, HarborError> {
    let parts: Vec<&str> = token.split('.').collect();
    if parts.len() != 3 {
        return Err(HarborError::Malformed);
    }
    let msg = format!("{}.{}", parts[0], parts[1]);
    let mut sig_bytes = Self::internal_decode_b64(parts[2])?;
    let sig_result = self.internal_verify_sig(
        msg.as_bytes(), &sig_bytes);
    sig_bytes.zeroize();  // Scrub signature from memory
    sig_result?;
    let mut payload_bytes = Self::internal_decode_b64(parts[1])?;
    let claims: HarborCardClaims =
        serde_json::from_slice(&payload_bytes)?;
    payload_bytes.zeroize();  // Scrub payload from memory
    if claims.exp < now_ts {
        return Err(HarborError::Expired);
    }
    Ok(claims)
}

pub fn constant_time_compare(a: &[u8], b: &[u8]) -> bool {
    if a.len() != b.len() { return false; }
    let mut result = 0u8;
    for i in 0..a.len() { result |= a[i] ^ b[i]; }
    result == 0
}
```

Listing 5: Deployed Rust Implementation — Harbor Card Verifier (abridged from the `harbor-card-rs` crate)

Key implementation details: `zeroize` requests that cryptographic material be overwritten when dropped, reducing residual-secret exposure without proving that no copy survives. The comparator avoids source-level early return on secret byte values. That is useful hygiene, not a hardware constant-time guarantee; compiler, cache, and microarchitectural behavior remain outside this source argument.

6.2 FFI Bridge to the Node.js Daemon

The Rust crate exports two C-ABI functions consumed by the TypeScript daemon:

```rust
#[no_mangle]
pub extern "C" fn harbor_constant_time_compare(
    a: *const u8, a_len: usize,
    b: *const u8, b_len: usize
) -> bool { /* ... */ }

#[no_mangle]
pub extern "C" fn harbor_verify_caps_subset_json(
    root_json: *const c_char, root_len: usize,
    sub_json: *const c_char, sub_len: usize
) -> bool { /* ... */ }
```

Listing 6: FFI Exports — Called from the daemon's Arbiter module

The daemon's `arbiter` module binds these at startup:

```typescript
constantTimeCompare: lib.func(
  'bool harbor_constant_time_compare(
     const uint8_t *a, size_t a_len,
     const uint8_t *b, size_t b_len)'),
verifyCapsSubset: lib.func(
  'bool harbor_verify_caps_subset_json(
     const char *root_json, size_t root_len,
     const char *sub_json, size_t sub_len)')
```

Listing 7: TypeScript FFI Binding (daemon-side Arbiter)

6.3 Kani Verification Harnesses

Three bounded model checking proofs are defined in the same source file under `#[cfg(kani)]`:

```rust
#[cfg(kani)]
#[kani::proof]
#[kani::stub(HarborCardVerifier::internal_pk_from_bytes,
             stubs::pk_from_bytes_stub)]
#[kani::stub(HarborCardVerifier::internal_decode_b64,
             stubs::decode_b64_stub)]
#[kani::stub(HarborCardVerifier::internal_verify_sig,
             stubs::verify_sig_stub)]
#[kani::unwind(10)]
fn proof_verify_logic_only() {
    let pk_bytes: [u8; 32] = kani::any();
    if let Ok(verifier) = HarborCardVerifier::new(pk_bytes) {
        let token_bytes: [u8; 32] = kani::any();
        if let Ok(token_str) =
            std::str::from_utf8(&token_bytes) {
            kani::assume(token_str.contains('.'));
            let _ = verifier.verify(token_str, 0);
        }
```

```
    }
}

#[cfg(kani)]
#[kani::proof]
fn proof_constant_time_behavior() {
    let a: [u8; 16] = kani::any();
    let b: [u8; 16] = kani::any();
    let _ = HarborCardVerifier::constant_time_compare(&a, &b);
}

#[cfg(kani)]
#[kani::proof]
fn proof_capability_attenuation() {
    let root = vec!["read".to_string(), "write".to_string()];
    let mut sub = vec!["read".to_string()];
    assert!(HarborCardVerifier::verify_capability_subset(
        &root, &sub));
    sub.push("admin".to_string());
    assert!(!HarborCardVerifier::verify_capability_subset(
        &root, &sub));
}
```

Listing 8: Kani Proof Harnesses (Deployed Source)

The stubbing strategy deserves explanation: `proof_verify_logic_only` stubs out Ed25519 signature verification and base64 decoding (which involve curve arithmetic that Kani cannot symbolically execute within practical bounds) and exhaustively explores the *parsing and control flow logic*—the split-on-dots, payload extraction, expiry comparison, and error handling paths. This verifies that the logic surrounding the cryptographic primitives never panics, never accesses out-of-bounds memory, and handles all malformed inputs gracefully, regardless of what the cryptographic layer returns.

The build output at `target/kani/aarch64-apple-darwin/debug` contains the CBMC intermediate representations used by Kani's bounded model checker.

7 Security Analysis

7.1 Threat Model

We assume a **Dolev-Yao adversary** [6] who:

- Controls the entire network
- Can intercept, modify, and inject messages
- Cannot break cryptographic primitives (cryptography is perfect)
- May corrupt legitimate agents (but not all simultaneously)

This is the standard threat model for symbolic protocol analysis [1].

7.2 Security Properties

Property	**Verified**	**Tool**	**Deployed**	**Reference**
Master Key Secrecy	Yes	ProVerif	—	[1]
Algorithm Confusion Immunity	Yes	ProVerif	—	[10]
Authentication correspondence	Yes	ProVerif	—	[5]
Delegation Integrity	Yes	ProVerif	—	[4]
Memory Safety (parsing)	Yes	Kani	Yes (FFI)	[21]
Secret-independent source branching	Yes	Kani	Yes (FFI)	[23]
Capability Attenuation	Yes	Kani	Yes (FFI)	[4]

Table 2: Verified Security Properties. "Deployed" indicates the verified Rust code is called by the production daemon via FFI from the Arbiter module into the `harbor-card-rs` crate.

7.3 Limitations

1. **Delegation-chain depth.** The symbolic proof of delegation soundness (Table 3, "`chain-replay.pv`") is machine-checked at the chain depth realized in deployment (depth 3); it is *not* a proof for unbounded chain length. ProVerif's unbounded-session guarantee quantifies over the number of protocol *sessions*, not over the number of hops in a single delegation chain, which is a fixed structural bound in the modeled process. Extending the proof to unbounded depth (via recursive replication of the delegation process) is future work. The current depth bound is a *spawn-time* policy check (`assessDelegation` in `lib/tube-spawner-router.ts`, `HARD_MAX_DELEGATION_DEPTH` = 8, default 4) applied when a spawn request arrives; it is *not* a verification-time invariant. The cryptographic chain verifier (`lib/delegation-chain.ts`) accepts arbitrary chain depth, and the attenuation monitor (`lib/cap-attenuation-monitor.ts`) enforces only per-hop capability monotonicity—not total depth. Machine-enforced depth at verification time is a planned hardening item.

2. **PID Binding:** The "Ghost in the Harbor" scenario requires binding tokens to the requesting PID, which is handled at the OS level.

3. **Local Transport Security:** On multi-user systems, local unix sockets or loopback TLS should be used to prevent local bearer token sniffing.

8 Conclusion

The Anchor Protocol replaces several hope-based assumptions with explicit, checkable obligations. It combines:

- Symbolic protocol proofs (ProVerif [1])
- Memory-safe implementation (Rust/Kani [21])
- Capability-based delegation (inspired by Macaroons [4])

Together these artifacts support a bounded assurance case for the modeled and tested surfaces. They do not certify the entire daemon, every delegation depth, or every side channel.

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A Formal Verification Status

Artifact availability. Every artifact named in Table 3 is bundled at `https://portdaddy.dev/whitepaper/artifacts/`. The verified Rust core is the open-source `harbor-card-rs` crate; runtime components are deployed inside the Port Daddy daemon binary. Chapter VI, *The Bonded Commons*, carries the cross-paper status table; this appendix lists only the items that pertain specifically to the Anchor Protocol.

Table 3: Anchor Protocol verification status. **closed**: model verified *and* runtime conformance test passes. PARTIAL: design verified, runtime empirically tested rather than proved. *open*: known unmechanized.

Claim	**Method**	**Result**	**Artifact**
Authentication correspondence in the listed phase models (§5)	ProVerif 2.05	**closed** non-injective correspondence queries TRUE	`harbor_card_v*.pv`
Algorithm-confusion immunity	ProVerif pinned vs. naive	**closed** pinned TRUE; counter-trace witnessed	`algconfusion.pv`
Delegation chain replay (§5.3)	ProVerif at depth 3	**closed** chain authenticity TRUE; 17-case runtime conformance	`chain-replay.pv`
Cuckoo-filter pollution resistance (§2.4)	5-invariant property test	**closed** FP rate within $2B/2^f$ at load 0.95	cuckoo property tests
GitHub-App webhook origin authentication	ProVerif (Dolev-Yao relay)	**closed** origin auth TRUE; daemon re-verifies GitHub HMAC (runtime conformance)	`ingress_origin_auth.pv`
Multi-tenant relay namespace isolation	ProVerif + runtime	**closed** member of A cannot publish into B; `HARBOR_MISMATCH` conformance	`ingress_tenant_isolation.pv`
Cuckoo-filter revocation soundness	inspection + empirical	PARTIAL gate narrows acceptance; dissemination remains assumption-bound	runtime test
Constant-time signature comparison	audited `subtle::ConstantTimeEq` library	PARTIAL no side-channel proof; library trust documented	—
Relay payload secrecy via HPKE (daemon key pinned)	ProVerif (design)	PARTIAL secrecy TRUE under pinned key; seal not yet implemented	`ingress_hpke_secrecy.pv`
Webhook replay-freedom + in-order delivery	TLA+ / TLC (exhaustive at bound)	PARTIAL design model-checked; runtime via ordered-chain ingest	`WebhookDelivery.tla`
Card revocation finality under backup/restore (ADR-0018 Attack 2)	TLA+ / TLC	PARTIAL rollback attack + revocation-epoch mitigation both model-checked	`CardRevocation.tla`
Mechanized gossip-convergence bound	—	*open* standard expected asymptotics only [26]; no partition deadline	—

Limitations. The transition to gossiping the Append-Only Event Log solves the bitwise merging impossibility, but cross-peer gossip propagation still leans on the standard anti-entropy analysis rather than a mechanized proof. The side-channel resistance of signature comparison is inherited from an audited library rather than re-proved. These deferrals are documented as known open problems rather than as defects in the present claims.

GitHub-App relay ingress. The webhook-dispatch surface (GitHub → untrusted relay → daemon) is modeled with the relay as the Dolev-Yao attacker, in three relay-agnostic properties, each paired with a negative control (`*_vuln.pv`) that exhibits the attack once the mitigation is removed—so a *true* result is non-vacuous. (i) *Origin authentication*: the daemon dispatches a fleet event only if GitHub signed it, even when the forwarder's transport credential is attacker-held; the control omits the daemon-side HMAC re-verification and the property fails (forged dispatch). This is *non-injective* agreement (origin), not replay-freedom: a captured genuine (*payload*, *mac*) may be re-delivered. Replay-freedom is an ordering-sensitive, mutable-state property outside ProVerif's reach, so it is discharged separately in TLA+ (`proofs/relay/WebhookDelivery.tla`): a daemon consuming the *ordered* per-publisher chain (dispatch iff $seq = tip + 1$) is model-checked at-most-once *and* gap-free against an adversarial relay that reorders, duplicates, and redelivers, with the unguarded daemon as the model-checked counterexample (TLC exhibits a double-dispatch). The obligation it names: the daemon ingests webhooks over the relay's ordered `from_seq` subscribe path, not the raw forward. (ii) *Payload secrecy*: under HPKE to the daemon's *pinned* X25519 key the relay never learns plaintext; the control sources the key from the relay and the secrecy property collapses to a key-substitution MITM—hence out-of-band key pinning is a requirement, not a recommendation. The seal is specified but not yet implemented, so this row is PARTIAL. (iii) *Namespace isolation*: a member of tenant A cannot publish into tenant B because harbor binding is checked against authoritative membership; the control trusts the card's self-declared issuer and crosses the tenant boundary. Honesty rider: HMAC-SHA256 gives origin authentication to a shared-key holder (EUF-CMA), not third-party non-repudiation; HPKE base mode is assumed IND-CCA2.

> *We err toward listing fewer items as closed rather than more.*

B Full ProVerif Models

For completeness, we include the full ProVerif models used for verification. These correspond to the algorithms in Section 5.

B.1 Phase 1: Symmetric HS256

```
(* Port Daddy: Harbor Card v1 (HS256) Formal Model *)
type id. type key. type jti. type capability. type alg_type.
const hs256_alg: alg_type. const none_alg: alg_type.
free c: channel.

fun hs256(bitstring, key): bitstring.
reduc forall m: bitstring, k: key;
  check_hs256(m, k, hs256(m, k)) = true.

reduc forall m: bitstring, k: key;
  verify_generic(m, hs256_alg, hs256(m, k), k) = true;
  forall m: bitstring, k: key;
  verify_generic(m, none_alg, m, k) = true.

fun encode_token(id, id, jti, capability): bitstring [data].

event Issued(id, id, jti, capability).
event Accepted(id, id, jti, capability).

free master_key: key [private].
query attacker(master_key).

query a: id, h: id, j: jti, cap: capability;
```

```
  event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).

let Daemon(k: key, a_id: id, h_id: id, j: jti, cap: capability) =
  let msg = encode_token(a_id, h_id, j, cap) in
  let signature = hs256(msg, k) in
  event Issued(a_id, h_id, j, cap);
  out(c, (hs256_alg, msg, signature)).

let SecureHarbor(k: key, expected_h: id) =
  in(c, (alg_header: alg_type, msg: bitstring, signature: bitstring))
     ;
  if check_hs256(msg, k, signature) = true then
    let encode_token(a: id, h: id, j: jti, cap: capability) = msg in
    if h = expected_h then
      event Accepted(a, h, j, cap).

process
  new agent_id: id; new harbor_id: id;
  new token_jti: jti; new secret_cap: capability;
  (!Daemon(master_key, agent_id, harbor_id, token_jti, secret_cap) |
   !SecureHarbor(master_key, harbor_id))
```

Listing 9: `harbor_card_v1_refined.pv`

C Phase 2: Asymmetric Ed25519

```
(* Port Daddy: Harbor Card v2 (Ed25519 / EdDSA) Formal Model *)
type id. type skey. type pkey. type jti. type capability.

free c: channel.
free k_dist: channel [private]. (* Trusted key distribution *)

(*** Digital Signatures (Ed25519) ***)
fun pk(skey): pkey.
fun sign(bitstring, skey): bitstring.
reduc forall m: bitstring, k: skey;
  check_sign(m, pk(k), sign(m, k)) = true.

fun encode_token(id, id, jti, capability): bitstring [data].

(*** Events ***)
event Issued(id, id, jti, capability).
event Accepted(id, id, jti, capability).

(*** Queries ***)
free secret_key: skey [private].
query attacker(secret_key).

query a: id, h: id, j: jti, cap: capability;
  event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).

(*** Processes ***)
let Daemon(a_id: id, h_id: id, j: jti, cap: capability) =
  in(k_dist, sk_h: skey);
  let msg = encode_token(a_id, h_id, j, cap) in
  let signature = sign(msg, sk_h) in
  event Issued(a_id, h_id, j, cap);
```

```
  out(c, (msg, signature)).

let Harbor(pk_h: pkey, expected_h: id) =
  in(c, (msg: bitstring, signature: bitstring));
  if check_sign(msg, pk_h, signature) = true then
    let encode_token(a: id, h: id, j_1: jti, cap_1: capability)
      = msg in
    if h = expected_h then
      event Accepted(a, h, j_1, cap_1).

(*** Main Process ***)
process
  new agent_id: id; new harbor_id: id;
  new token_jti: jti; new secret_cap: capability;
  let harbor_pk = pk(secret_key) in
  out(c, harbor_pk);
  (
    (!Daemon(agent_id, harbor_id, token_jti, secret_cap)) |
    (!Harbor(harbor_pk, harbor_id)) |
    (!out(k_dist, secret_key))
  )
```

Listing 10: `harbor_card_v2_asymmetric.pv` — Verified in ProVerif 2.05

D Phase 3: Multi-hop Delegation

```
(* Port Daddy: Harbor Card v3 (Delegation Chains) Formal Model *)
type id. type skey. type pkey. type capability.

free c: channel.

(*** Cryptographic Functions ***)
fun pk(skey): pkey.
fun sign(bitstring, skey): bitstring.
reduc forall m: bitstring, k: skey;
  check_sign(m, pk(k), sign(m, k)) = true.

reduc forall c1: capability; is_subset(c1, c1) = true.

fun base_token(id, id, id, capability): bitstring [data].
fun delegate_token(bitstring, id, capability): bitstring [data].

(*** Events ***)
event IssuedRoot(id, id, capability).
event Delegated(id, id, capability).
event Accepted(id, id, capability).

(*** Queries ***)
free harbor_id: id.
query b: id, cap: capability, a: id;
  event(Accepted(b, harbor_id, cap)) ==>
    (event(IssuedRoot(a, harbor_id, cap))
     && event(Delegated(a, b, cap)))
    || event(IssuedRoot(b, harbor_id, cap)).

(*** Processes ***)
free daemon_id: id.
```

```
free daemon_sk: skey [private].

let Daemon(a: id, cap: capability) =
  let msg = base_token(daemon_id, a, harbor_id, cap) in
  let s = sign(msg, daemon_sk) in
  event IssuedRoot(a, harbor_id, cap);
  out(c, (msg, s)).

let AgentA(a: id, a_sk: skey, b: id, sub_cap: capability) =
  in(c, (msg: bitstring, s: bitstring));
  let base_token(=daemon_id, =a, =harbor_id,
    root_cap: capability) = msg in
  if check_sign(msg, pk(daemon_sk), s) = true then
    if is_subset(sub_cap, root_cap) = true then
      let d_msg = delegate_token((msg, s), b, sub_cap) in
      let d_s = sign(d_msg, a_sk) in
      event Delegated(a, b, sub_cap);
      out(c, (d_msg, d_s)).

let Harbor(a_pk: pkey) =
  in(c, (d_msg: bitstring, d_s: bitstring));
  let delegate_token((base_msg: bitstring, base_s: bitstring),
    b: id, sub_cap: capability) = d_msg in
  let base_token(=daemon_id, a: id, =harbor_id,
    root_cap: capability) = base_msg in
  if check_sign(base_msg, pk(daemon_sk), base_s) = true then
  if check_sign(d_msg, a_pk, d_s) = true then
  if is_subset(sub_cap, root_cap) = true then
    event Accepted(b, harbor_id, sub_cap).

(*** Main Process ***)
process
  new sk_a: skey;
  let pk_a = pk(sk_a) in
  out(c, pk_a);
  new id_a: id; new id_b: id; new cap_root: capability;
  (
    (!Daemon(id_a, cap_root)) |
    (!AgentA(id_a, sk_a, id_b, cap_root)) |
    (!Harbor(pk_a))
  )
```

Listing 11: `harbor_card_v3_delegation.pv` — reflexive-order model (see §D.1 for the soundness caveat and the enriched v5 proof)

D.1 Soundness of the Attenuation Proof (v5)

The v3 model above is honest about cryptographic structure but *vacuous about attenuation*. Its capability order is reflexive only:

```
reduc forall c1: capability; is_subset(c1, c1) = true.
```

Capabilities are opaque atoms with no order, so a delegated capability can only ever *equal* its parent. Escalation—delegating more authority than you hold—is not blocked; it is *inexpressible*. The no-escalation query therefore holds for a reason that has nothing to do with the protocol: there is no larger capability to delegate. A reviewer flagged this directly: *"the model cannot witness escalation."*

`harbor_card_v5_attenuation.pv` closes the obligation by enrichment, not by weakening the claim. It gives capabilities a real partial order (`cap_read` $\sqsubset$ `cap_write`, both public so the attacker can construct either), replaces the reflexive stub with the sound order (`is_subset(cap_read, cap_write)` derivable; `is_subset(cap_write, cap_read)` *not*), and adds an *insider escalation adversary*: an agent holding a valid `cap_read` root token that signs an over-broad `cap_write` delegation with no self-restraint. The verifier's `is_subset` guard is the sole defense under test.

```
(* Sound order: reflexive + read <= write; write <= read NOT derivable *)
fun is_subset(capability, capability): bool
reduc
    forall x: capability; is_subset(x, x) = true
    otherwise is_subset(cap_read, cap_write) = true.

(* Insider adversary: holds a real root, signs an over-broad delegation *)
let MaliciousA(a: id, a_sk: skey, b: id) =
  in(c, (msg: bitstring, s: bitstring));
  let base_token(=daemon_id, =a, =harbor_id, root_cap: capability) = msg in
  if check_sign(msg, pk(daemon_sk), s) = true then
    event EscalationAttempted(a, root_cap, cap_write);
    let d_msg = delegate_token((msg, s), b, cap_write) in
    out(c, (d_msg, sign(d_msg, a_sk))).

(* Q1 soundness: a write-card delegated from a read-root is NEVER accepted *)
query b: id; event(Accepted(b, harbor_id, cap_write, cap_read)) ==> false.
(* Q2 non-vacuity: the escalation IS reachable (the proof is not vacuous) *)
query a: id, r: capability, s: capability;
  event(EscalationAttempted(a, r, s)) ==> false.
```

Listing 12: `harbor_card_v5_attenuation.pv` (excerpt) — Verified in ProVerif 2.05

Theorem D.1 (Attenuation soundness, mechanized)**.** In `harbor_card_v5_attenuation.pv`, ProVerif 2.05 reports `Accepted(b, harbor, cap_write, cap_read)` unreachable (**Q1 = true**): no escalating delegation is ever accepted. Simultaneously `EscalationAttempted` is reachable (**Q2 = false-with-trace**): the attack is expressible, so Q1 is a real defense, not the v3 vacuum. A negative control that deletes the verifier's `is_subset` guard flips Q1 to false-with-trace—ProVerif exhibits the read→write escalation— confirming the guard is load-bearing.

Multi-hop scope (honest boundary). Theorem D.1 is single-hop (Daemon $\to A \to$ verifier). Multi-hop delegation has a sharper obligation: a verifier that checks the *final* capability against the *root* accepts a non-monotonic middle hop. `harbor_card_v6_multihop_attack.pv` mechanizes exactly this — A:`write` $\to$ B:`read` $\to$ C:`write` is accepted (ProVerif: escalation reachable). The fix, `harbor_card_v7_multihop_fixed.pv`, verifies each hop against its *immediate parent* (is_subset(*final*, *mid*) $\wedge$ is_subset(*mid*, *root*)) and ProVerif proves the same chain rejected. The runtime delegation verifier — and the cross-harbor recompute $\mathrm{att}_B \circ \mathrm{att}_A$ — must be per-hop, never final-vs-root.

E Limitations & Boundaries

The formal mechanisms presented in this chapter are mathematically sound within their defined scopes, but they depend on bounding assumptions that must be explicitly acknowledged.

- **Threat Model Exclusions:** Collusion across all independent organs (e.g., the logging daemon, the arbitrator, and the primary execution runtime) is assumed out-of-scope; if all enforcing components are compromised, the guarantees degrade to advisory.
- **Performance Trade-offs:** Cryptographic assertions and WAL-checkpointing for per-write durability incur measurable I/O overhead. These mechanisms are designed for high-stakes coordination, not for bulk data processing or high-frequency trading.
- **Implementation Maturity:** While the core protocols (Anchor, SQLite single-writer) are IMPLEMENTED and running in production, surrounding ecosystem components (like the decentralized judge markets or fully federated multi-sig routing) remain in PARTIAL or DESIGNED phases.

These constraints are not flaws but structural realities of building zero-trust coordination on top of POSIX primitives.